

- 5 **D** By applying PCM in East-West direction,
The total momentum of system before and after multiple collisions should be equal to zero.
Thus, after the ball has exited the box, momenta in East-West direction of both box and ball will be of opposite directions.
If the ball moves east, the box will move west.
Only option D is possible.

- 6 **D** $n_1 > n_2$